Radioactive emission is a fundamentally quantum process. What we informally describe as the occasional, random emission of a particle is actually a smooth evolution of the wave function of the atomic nuclei within the source. Each nucleus evolves from a state of purely un-decayed to a superposition of (un-decayed)+(decayed), with the latter part gradually growing over time. The emission appears random because the detector doesn't measure the wave function directly; it only sees either (un-decayed) or (decayed), just as a vertical Stern-Gerlach magnet only ever sees spin-up or spin-down.

The point of the thought experiment is to take a microscopic quantum superposition and magnify it to a manifestly macroscopic situation. That happens as soon as the detector clicks. All the business with the sleeping gas and the cat is just to make the amplification of a quantum superposition to the macroscopic world more vivid. (The word "entanglement," or in German *Verschränkung*, was first applied to quantum mechanics by Schrödinger in the discussion of his cat, which arose out of correspondence with Einstein.)

Schrödinger's experiment was posed in the context of the textbook approach to the measurement problem, where wave functions collapse when they are literally observed. So, he says, imagine that we keep the box closed—not observing what's inside—until the wave function evolves to an even superposition of "at least one nucleus has decayed" and "no nuclei have decayed." In that case, the wave functions of the detector, the gas, and the cat will all also evolve into an equal superposition, of "the detector clicked, the gas was released, and the cat is asleep" and "the detector didn't yet click, the gas is still in the container, and the cat is awake." Surely, asks Schrödinger, you don't seriously believe that the box contains a superposition of an awake cat and an asleep cat until we open it?